

ICS 6B: Homework 1

1.1.2

(a)

$$n \wedge m$$

(b)

$$t \wedge m$$

(c)

$$n \vee m$$

(d)

$$\neg m$$

(e)

$$t \wedge n$$

(f)

$$\neg t$$

1.1.4

(d)

Inclusive or: True

Exclusive or: False

1.2.4

(d)

p	q	r	$\neg r$	$\neg q$	$(r \vee p)$	$(\neg r \vee \neg q)(\neg r \vee \neg q)$	$(r \vee p) \wedge (\neg r \vee \neg q)$
F	F	F	T	T	F	T	F
F	F	T	F	T	T	T	T
F	T	F	T	F	F	T	F
F	T	T	F	F	T	F	F
T	F	F	T	T	T	T	T
T	F	T	F	T	T	T	T
T	T	F	T	F	T	T	T
T	T	T	F	F	T	F	F

1.2.7

(c)

$$B \vee (D \wedge M)$$

1.3.3

(b)

1. Converse

1. Statement: If $5 < 3$, then $7 < 5$.

2. $\text{False} \rightarrow \text{False}$.

3. True

2. Inverse

1. Statement: If 7 is not less than 5, then 5 is not less than 3

2. $\text{True} \rightarrow \text{True}$

3. True

3. Contrapositive

1. Statement: If 5 is not less than 3, then 7 is not less than 5.

2. $\text{True} \rightarrow \text{True}$

3. True

1.3.7

(a)

they are a senior and at least 17 years of age = $s \wedge y$

Thus $p \rightarrow s \wedge y$

(c)

$p \rightarrow y$

1.3.10

(b)

$$(T \vee r) \rightarrow r$$

Left side is $T \rightarrow r$

This is **Unknown**

(c)

$$(T \vee r) \leftrightarrow (F \wedge r)$$

Left side is $T \vee r = T$

Right side is $F \wedge r = F$

Thus $T \leftrightarrow F$

This is **False**

(e)

$$T \rightarrow (r \vee F)$$

Right side is $r \vee F = r$

Thus $T \rightarrow r$

This is **Unknown**

(f)

$$(T \wedge F) \rightarrow r$$

Left side is

$$T \wedge F = F$$

Thus $F \rightarrow r$

This is **True**

1.4.4

(a)

p	q	$\neg q$	$p \vee \neg q$	$\neg(p \vee \neg q)$	$\neg p$	$\neg p \wedge q$
T	T	F	T	F	F	F
T	F	T	T	F	F	F
F	T	F	F	T	T	T
F	F	T	T	F	T	F

The columns for $\neg(p \vee \neg q)$ and $\neg p \wedge q$ are identical. Therefore, they are logically equivalent.

(c)

p	q	$\neg q$	$p \vee \neg q$	$\neg(p \vee \neg q)$	$\neg p$	$\neg q$	$\neg p \wedge \neg q$
T	T	F	T	F	F	F	F
T	F	T	T	F	F	T	F
F	T	F	F	T	T	F	F
F	F	T	T	F	T	T	T

The columns for $p \wedge (p \rightarrow q)$ and $p \wedge q$ are identical. Therefore, they are logically equivalent.

1.4.5

(b)

Expression 1 is $\neg j \rightarrow (l \vee \neg r)$

Expression 2 is $(r \wedge \neg l) \rightarrow j(r \wedge \neg l) \rightarrow j$

j	l	r	$\neg j$	$\neg r$	$l \vee \neg r$	Expr 1: $\neg j \rightarrow (l \vee \neg r)$	$\neg l$	$r \wedge \neg l$	Expr 2: $(r \wedge \neg l) \rightarrow j$
T	T	T	F	F	T	T	F	F	T
T	T	F	F	T	T	T	F	F	T
T	F	T	F	F	F	T	T	T	T
T	F	F	F	T	T	T	T	F	T
F	T	T	T	F	T	T	F	F	T
F	T	F	T	T	T	T	F	F	T
F	F	T	T	F	F	F	T	T	F
F	F	F	T	T	T	T	T	F	T

Since the result of both expressions are the same, they are equivalent.

(d)

Expression 1 is $j \rightarrow \neg l$

Expression 2 is $\neg j \rightarrow l$

j	l	$\neg l$	Expr 1: $j \rightarrow \neg l$	$\neg j \neg j$	Expr 2: $\neg j \rightarrow l$
T	T	F	F	F	T
T	F	T	T	F	F
F	T	F	T	T	T
F	F	T	T	T	F

Since the result of both expressions are the not same, they are not equivalent.

1.5.1

(c)

1. $r \vee (\neg r \rightarrow p)$
2. $r \vee (\neg \neg r \vee p)$ **Conditional identities**
3. $r \vee (r \vee p)$ **Double Negation Law**
4. $(r \vee r) \vee p$ **Associative Law**
5. $r \vee p$ **Idempotent Law**

1.5.2

(c)

Given $(p \rightarrow q) \wedge (p \rightarrow r)$

Implication law $(\neg p \vee q) \wedge (\neg p \vee r)$

Distributive law $\neg p \vee (q \wedge r)$

Implication law $p \rightarrow (q \wedge r)$

(f)

Given $\neg(p \vee (\neg p \wedge q))$

De Morgans law $\neg p \wedge (\neg(\neg p) \vee \neg q)$

Double negative law $\neg p \wedge (p \vee \neg q)$

Distributive law $(\neg p \wedge p) \vee (\neg p \wedge \neg q)$

Negation law $F \vee (\neg p \wedge \neg q)$

Identity law $\neg p \wedge \neg q$

1.5.4

(b)

Let $p \rightarrow q$

The inverse is $\neg p \rightarrow \neg q$

Condition:

Implication law $\neg p \vee q$

Inverse:

Implication law

$\neg(\neg p) \vee \neg q$

Double negation law $p \vee \neg q$

Thus, they are not logically equivalent.

(c)

Let $p \rightarrow q$

The contrapositive is $\neg q \rightarrow \neg p$

Condition:

Implications law $\neg p \vee q$

Commutative law $q \vee \neg p$

Double Negation law $\neg(\neg q) \vee \neg p$

Implication Law

$\neg q \rightarrow \neg p$

Thus, conditional statement is logically equivalent to its contrapositive.